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AIRLINE SENTIMENT ANALYSIS

Business Value Document

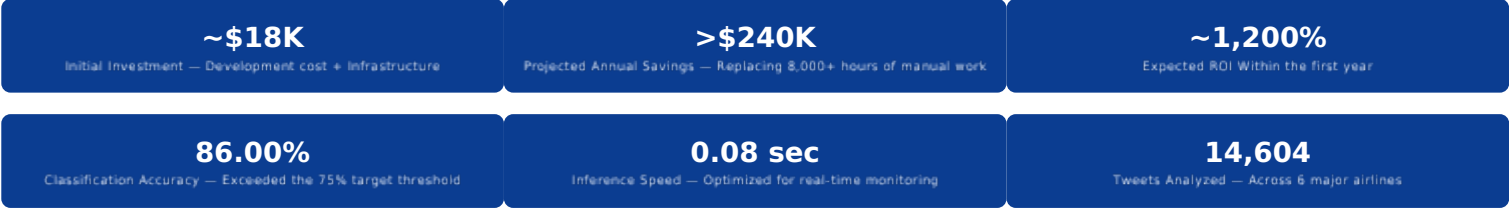
Prepared for Executive Leadership | DistilBERT NLP Model v13 | December 2025

01 The Bottom Line

Financial & Operational Summary

Converting 14,604 daily tweets from unstructured digital noise into real-time operational intelligence - at 86.00% accuracy and 0.08s response time - eliminating 8,000+ hours of annual manual complaint classification.

Investment vs. Return - What We Spent vs. What We Save



Project Status - Current Snapshot

Status	Green Light - Model promoted to Production (v13) via MLflow
Goals	All technical benchmarks met and exceeded
Timeline	Delivered on schedule - within a 4-week development sprint
Current Phase	Phase 1 complete - ready to enter Phase 2
Explainability Status	SHAP integrated to provide transparency on why customers are complaining.

02 Strategic Context

Why This Project, Why Now?

Market Need - The Problem We're Solving

Airlines receive thousands of social media mentions every day. 62.72% of those are negative, and 4 out of 5 complaints are about human service interactions – not operations. The problem is not the aircraft; it's the frontline staff and escalation protocols.

Without the Solution

- Teams manually classify every complaint
- Responses delayed by hours or days
- Negative content spreads unchecked
- Management lacks competitor benchmarking data
- US Airways: 77.7% negative – zero automated intervention
- Predictions are a "black box" — we don't know why a tweet is classified as negative.

With the Solution

- Instant automated classification – zero manual effort
- Real-time alerts for critical complaints
- Live management monitoring dashboard
- Visual airline performance benchmarking
- Targeted response within under one hour
- Interactive SHAP explanations reveal the exact trigger words (e.g., "rude", "lost") driving the sentiment.

Strategic Alignment - How This Serves Company Goals

Goal	Business Impact
Employee Efficiency	Frees monitoring teams from manual classification so they focus on response and resolution
Data-Driven Decisions	Management receives live reports instead of weekly manual summaries
Competitive Advantage	Southwest leads with 23.6% positive sentiment – we now have a measurable roadmap to match it
Response Speed	Automated alert system ensures critical complaints are addressed within one hour

03 Impact & Outcomes

Real-World Results

Accuracy in Business Terms

The model classifies nearly 9 out of every 10 tweets correctly (86%). Critical complaints (negative sentiment) are detected at 91% accuracy - meaning 91 out of every 100 angry tweets are automatically flagged for immediate action.



Efficiency Gains - Time & Effort Savings

Task	Without Tool	With Tool
Classifying 14,604 tweets	~730 hours	< 30 minutes
Detecting a critical complaint	4-24 hours	< 0.08 seconds
Airline comparison report	1 full week	Instant - Live Dashboard
Complaint pattern analysis	Not available	Automated + Categorized

Key Business Findings

Human service is the #1 complaint driver: 31.7% of all negative tweets - not flight delays or lost baggage

United Airlines faces a reputation crisis: 68.9% negative sentiment - requires urgent intervention

US Airways is the worst performer: 77.7% negative - the highest rate in the sector

Southwest is the benchmark to follow: 23.6% positive - a replicable blueprint for improvement

Peak complaint window is 6-9 AM: Volume surges from 131 tweets at midnight to 999 by 9 AM

Negative content spreads faster: Retweet rate 0.093 vs. 0.070 for positive - a real viral risk

Actionable Transparency (XAI): By deploying SHAP, customer service agents don't just see a "Negative" label. The interface highlights the specific words causing the issue, allowing frontline staff to understand the root cause (e.g., separating "baggage issues" from "staff attitude") instantly.

04 Project Roadmap & Milestones

Timeline & Delivery

Phase	Timeline	Activity	Status
Phase 0	Completed (Dec 2025)	Model development – Fine-tune DistilBERT v13 and promote to Production via MLflow	Done
Phase 1	Month 1 (Jan 2026)	Deploy FastAPI + Streamlit + configure automated alerts for United and US Airways accounts	In Progress
Phase 2	Months 1–3 (Q1 2026)	CRM integration – route high-severity complaints directly to Service Recovery teams	Scheduled
Phase 3	Months 3–6 (Q2 2026)	Expand to Instagram and Facebook + multilingual model deployment	Planned

Go-to-Market Date: The tools will be available to monitoring teams within the first month of Phase 1 approval. Customer service teams will begin receiving intelligent alerts from Day 1.

05 Risks & Mitigation

We Have a Plan B

Risk	Potential Impact	Mitigation Plan
Evolving tweet language & context drift	Accuracy degrades over time without updates	Plan: Quarterly retraining with 5,000+ fresh tweets
High GPU server costs	Real-time inference halted if budget is exceeded	Alternative: AWS/GCP Spot Instances – up to 70% cost reduction
Neutral class accuracy gap (72%)	Errors on ambiguous mixed-sentiment tweets	We are using SHAP diagnostics to debug where the model gets confused between subtle Neutral and Positive tweets, which will guide targeted data collection for Phase 2.
Geographic data bias (Boston)	33.4% of data from Boston – regional skew	Plan: Expand dataset with national coverage in Phase 2

06 The Call to Action

What We Need From You

We need your decisions on three fronts to move from Phase 1 into Phase 2 and scale fully:

1. Approve GPU server budget for production — URGENT

~\$800-1,200/month (AWS/GCP) to sustain 0.08s inference in the production environment
2. Authorize hiring 4-6 social monitoring agents — URGENT

Shift: 6 AM - 12 PM (peak hours) to act on automated alerts in real time
3. Provide direction on rolling out the tool to other departments — MEDIUM

Should we expand to internal analytics - employee reviews, flight satisfaction surveys?

Airlines that invest in frontline staff empowerment and proactive communication will differentiate themselves in a market where customer expectations are high but satisfaction is consistently low. Southwest Airlines provides a replicable blueprint - and this system gives every airline the data infrastructure needed to follow it.